Mack's chain-ladder model in Lean 4: blueprint

Robert Sneiderman

August 25, 2026

Contents

Chapter 1

Introduction

This blueprint tracks the formalization of Mack's distribution-free chain-ladder model [1] in Lean 4 on top of mathlib. Each statement below carries the name of its Lean declaration; green nodes in the dependency graph are proved, the others are open work listed in the roadmap. Statements are given in the paper's own terms, and where a published formula is an estimator or an approximation rather than an identity, the text says so.

A run-off triangle with n accident years and n development years is a function $C : \mathbb{N} \to \mathbb{N} \to \mathbb{R}$; $C_{i,k}$ is the cumulative claims of accident year i at development year k, observed when $i+k \leq n-1$. The latest observed entry of accident year i is $C_{i,\,n-1-i}$.

Chapter 2

The chain-ladder estimators

2.1 Definitions

Definition 2.1 (Contributors)**.** The accident years contributing to the development factor $k \to k+1$ are $i = 0, \dots, n-k-2$, those with $C_{i,k+1}$ observed.

Definition 2.2 (Column sums)**.** $S_k = \sum_{i=0}^{n-k-2} C_{i,k}$ and $T_k = \sum_{i=0}^{n-k-2} C_{i,k+1}$.

Definition 2.3 (Development factor)**.** $\hat{f}_k = T_k / S_k$. (In Lean, division by zero is zero.)

Definition 2.4 (Individual factor)**.** $F_{i,k} = C_{i,k+1}/C_{i,k}$.

Definition 2.5 (Mack's variance estimator)**.** $\hat{\sigma}_k^2 = \frac{1}{n-k-2} \sum_{i=0}^{n-k-2} C_{i,k} \, (F_{i,k} - \hat{f}_k)^2$. The value for $k = n-2$ (a single contributor) is an extrapolation convention in [1], not an estimator; no theorem here uses it.

Definition 2.6 (Projection)**.** $\hat{C}_{i,k} = C_{i,n-1-i} \prod_{j=n-1-i}^{k-1} \hat{f}_j$ for $k \geq n-1-i$.

Definition 2.7 (Ultimate and reserve)**.** $\hat{C}_{i,n-1}$ is the chain-ladder ultimate and $\hat{R}_i = \hat{C}_{i,n-1} - C_{i,n-1-i}$ the reserve.

Definition 2.8 (Mack's MSEP estimator)**.** Mack (1993), Theorem 3:

$$\widehat{\mathrm{msep}}(\hat{R}_i) = \hat{C}_{i,n-1}^2 \sum_{k=n-1-i}^{n-2} \frac{\hat{\sigma}_k^2}{\hat{f}_k^2} \Big(\frac{1}{\hat{C}_{i,k}} + \frac{1}{S_k} \Big).$$

2.2 Deterministic identities

Lemma 2.9 (Weighted-average form)**.** *If $C_{i,k} \neq 0$ for every contributor i, then $\hat{f}_k = \sum_i \frac{C_{i,k}}{S_k} F_{i,k}$.*

Proof. Termwise: $\frac{C_{i,k}}{S_k} \cdot \frac{C_{i,k+1}}{C_{i,k}} = \frac{C_{i,k+1}}{S_k}$. ☐

Lemma 2.10. *If $C_{i,k} \neq 0$ for every contributor, $T_k = \sum_i C_{i,k} F_{i,k}$.*

Proof. Cancel the denominator termwise. ☐

Lemma 2.11 (Sum-of-squares decomposition)**.** *For any centre f, with $C_{i,k} \neq 0$ for every contributor, $\sum_i C_{i,k}(F_{i,k} - f)^2 = \sum_i C_{i,k} F_{i,k}^2 - 2fT_k + f^2 S_k$.*

Proof. Expand the square and use Lemma 2.10. □

Lemma 2.12 (Collapse at the chain-ladder centre)**.** *If moreover $S_k \neq 0$, then $\sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2 = \sum_i C_{i,k}F_{i,k}^2 - S_k\hat{f}_k^2$. This is the identity behind the $n-k-2$ degrees of freedom in $\hat{\sigma}_k^2$.*

Proof. Put $f = \hat{f}_k = T_k/S_k$ in Lemma 2.11. □

Lemma 2.13 (Projection step)**.** *For $k \geq n-1-i$, $\hat{C}_{i,k+1} = \hat{C}_{i,k}\,\hat{f}_k$.*

Proof. Split off the top factor of the product. □

Lemma 2.14. $\hat{C}_{i,n-1-i} = C_{i,n-1-i}$.

Proof. Empty product. □

Lemma 2.15. *The oldest accident year is fully developed: $\hat{R}_0 = 0$.*

Proof. Empty product again. □

Chapter 3

Mack 1999 equals Mack 1993

Definition 3.1 (Mack's 1999 recursion)**.** For $\alpha = 1$ and unit weights [2], starting from $\mathrm{se}^2 = 0$ on the latest diagonal $d = n - 1 - i$ and stepping m times,

$$\mathrm{se}^2_{m+1} = \hat{C}^2_{i,d+m}\Big(\frac{\hat{\sigma}^2_{d+m}}{\hat{C}_{i,d+m}} + \frac{\hat{\sigma}^2_{d+m}}{S_{d+m}}\Big) + \mathrm{se}^2_m\,\hat{f}^2_{d+m}.$$

Definition 3.2. The summand of the closed form: $\mathrm{term}_{i,k} = \frac{\hat{\sigma}^2_k}{\hat{f}^2_k}\big(\frac{1}{\hat{C}_{i,k}} + \frac{1}{S_k}\big)$.

Lemma 3.3. $\widehat{\mathrm{msep}}(\hat{R}_i) = \hat{C}^2_{i,n-1}\sum_{k=n-1-i}^{n-2}\mathrm{term}_{i,k}$.

Proof. By definition. ◻

Lemma 3.4 (Truncated closed form)**.** *If $\hat{f}_k \neq 0$ for $d \le k < d{+}m$, then $\mathrm{se}^2_m = \hat{C}^2_{i,d+m}\sum_{k=d}^{d+m-1}\mathrm{term}_{i,k}$.*

Proof. Induction on m using Lemma 2.13 and $\hat{f}^2_{d+m}/\hat{f}^2_{d+m} = 1$. ◻

Theorem 3.5 (Mack 1999 = Mack 1993)**.** *For $i \le n-1$ and $\hat{f}_k \neq 0$ along the row, i steps of the 1999 recursion return exactly Mack's 1993 estimator: $\mathrm{se}^2_i = \widehat{\mathrm{msep}}(\hat{R}_i)$.*

Proof. Lemma 3.4 with $m = i$, since $d + i = n - 1$. ◻

Chapter 4

Bornhuetter-Ferguson on the chain-ladder pattern

Definition 4.1 (Cumulative development factor, BF reserve and ultimate)**.** $\mathrm{cdf}_i = \prod_{k=d}^{n-2} \hat{f}_k$; for an a priori ultimate U, $R_i^{BF} = U\,(1 - 1/\mathrm{cdf}_i)$ and $\hat{C}_{i,n-1}^{BF} = C_{i,d} + R_i^{BF}$.

Lemma 4.2 (Linearity)**.** *$R_i^{BF}(aU) = a\,R_i^{BF}(U)$ and $R_i^{BF}(U+V) = R_i^{BF}(U) + R_i^{BF}(V)$.*

Proof. Ring identities. □

Theorem 4.3 (BF with the chain-ladder prior is chain ladder)**.** *If $\mathrm{cdf}_i \neq 0$ and $U = \hat{C}_{i,n-1}$, then $\hat{C}_{i,n-1}^{BF} = \hat{C}_{i,n-1}$ and $R_i^{BF} = \hat{R}_i$.*

Proof. $\hat{C}_{i,n-1} = C_{i,d}\,\mathrm{cdf}_i$, so $U(1 - 1/\mathrm{cdf}_i) = C_{i,d}\mathrm{cdf}_i - C_{i,d}$. □

Lemma 4.4 (The BF reserve is a fraction of the prior)**.** *If every development factor along the row is at least 1, then $\mathrm{cdf}_i \geq 1$ and, for $U \geq 0$, $0 \leq R_i^{BF}(U) \leq U$.*

Proof. A product of numbers at least 1 is at least 1; then $0 \leq 1 - 1/\mathrm{cdf}_i \leq 1$. □

Chapter 5

The stochastic layer

5.1 Model

Definition 5.1 (Random triangle). A random run-off triangle on a measurable space Ω is a family of random variables $C_{i,k} : \Omega \to \mathbb{R}$ together with a filtration $\mathcal{D}_k$ such that $C_{i,j}$ is $\mathcal{D}_k$-measurable whenever $j \le k$ ("everything observed up to development year k").

Definition 5.2 (Assumption (M1)). $E[C_{i,k+1} \mid \mathcal{D}_k] = f_k\, C_{i,k}$ almost surely, for all i and k. This is Mack's first assumption in the form conditioned on $\mathcal{D}_k$; Mack states it conditioned on the row's own history and obtains this form from independence across accident years (M2). That passage is Theorem 5.30.

Definition 5.3 (Estimators as random variables). $S_k(\omega)$ and $\hat{f}_k(\omega)$ are the deterministic estimators applied to the triangle observed at ω.

Lemma 5.4. *S_k is $\mathcal{D}_k$-measurable.*

Proof. A finite sum of $\mathcal{D}_k$-measurable functions. □

5.2 Mack's Theorem 2

Theorem 5.5 (Conditional unbiasedness of the development factor). *Assume (M1), that $S_k \neq 0$ almost surely, that each $C_{i,k+1}$ and $\hat{f}_k$ are integrable. Then $E[\hat{f}_k \mid \mathcal{D}_k] = f_k$ almost surely.*

Proof. $\hat{f}_k = S_k^{-1} \sum_i C_{i,k+1}$. The factor S_k^{-1} is $\mathcal{D}_k$-measurable and comes out of the conditional expectation; by (M1) the conditional expectation of the sum is $f_k \sum_i C_{i,k} = f_k S_k$; cancel S_k where it is nonzero. □

Lemma 5.6. *$\hat{f}_j$ is $\mathcal{D}_k$-measurable whenever $j + 1 \le k$.*

Proof. S_j is $\mathcal{D}_j \subseteq \mathcal{D}_k$-measurable and each $C_{i,j+1}$ is $\mathcal{D}_k$-measurable. □

Theorem 5.7 (Uncorrelated development factors, conditional form). *Under (M1), on a finite measure space, for $j < k$ (with the integrability and $S \neq 0$ hypotheses of Theorem 5.5 for both columns and $\hat{f}_j \hat{f}_k$ integrable), $E[\hat{f}_j \hat{f}_k \mid \mathcal{D}_j] = f_j f_k$ almost surely.*

Proof. Condition on $\mathcal{D}_k$ first: $\hat{f}_j$ is $\mathcal{D}_k$-measurable and comes out, and $E[\hat{f}_k \mid \mathcal{D}_k] = f_k$ by Theorem 5.5. Then the tower property down to $\mathcal{D}_j$ and Theorem 5.5 again for column j. □

Theorem 5.8 (Mack 1993, Theorem 2). *On a probability space, under the same hypotheses, $E[\hat{f}_k] = f_k$ and $E[\hat{f}_j \hat{f}_k] = f_j f_k$ for $j < k$: the chain-ladder factors are unbiased and uncorrelated, which is what makes $\prod_k \hat{f}_k$ an unbiased estimator of $\prod_k f_k$.*

Proof. Integrate the conditional statements. □

5.3 Mack's Theorem 1

Definition 5.9 (Chain-ladder projection as a random variable). $\hat{C}_{i,d+m} = C_{i,d} \prod_{k=d}^{d+m-1} \hat{f}_k$ with $d = n - 1 - i$.

Lemma 5.10. *$\hat{C}_{i,d+m}$ is $\mathcal{D}_{d+m}$-measurable.*

Proof. Induction on m; each new factor $\hat{f}_{d+m}$ is $\mathcal{D}_{d+m+1}$-measurable. □

Lemma 5.11 (Iterating (M1)). *For $d \le k$, $E[C_{i,k+1} \mid \mathcal{D}_d] = f_k\, E[C_{i,k} \mid \mathcal{D}_d]$, hence $E[C_{i,d+m} \mid \mathcal{D}_d] = C_{i,d} \prod_{k=d}^{d+m-1} f_k$.*

Proof. Tower property through $\mathcal{D}_k$, then (M1); induction on m. □

Theorem 5.12 (Mack 1993, Theorem 1). *Under (M1), with the partial products integrable and the column sums a.s. nonzero, $E[\hat{C}_{i,d+m} \mid \mathcal{D}_d] = C_{i,d} \prod_{k=d}^{d+m-1} f_k$, and therefore $E[\hat{C}_{i,n-1} \mid \mathcal{D}_d] = E[C_{i,n-1} \mid \mathcal{D}_d]$: the chain-ladder ultimate is a conditionally unbiased estimator of the true ultimate claims.*

Proof. Condition on $\mathcal{D}_{d+m}$: the partial product comes out and the last factor has conditional expectation f_{d+m} (Theorem 5.5); tower down to $\mathcal{D}_d$ and induct. Compare with Lemma 5.11. □

5.4 The variance assumptions and the estimation variance of $\hat{f}_k$

Definition 5.13 (Residuals, (M3) and (M2')). With $\varepsilon_{i,k} = C_{i,k+1} - f_k C_{i,k}$: (M3) $E[\varepsilon_{i,k}^2 \mid \mathcal{D}_k] = \sigma_k^2 C_{i,k}$, and (M2') $E[\varepsilon_{i,k}\varepsilon_{j,k} \mid \mathcal{D}_k] = 0$ for $i \ne j$, the conditional uncorrelatedness across accident years that independence of accident years supplies.

Lemma 5.14. *On $\{S_k \ne 0\}$, $\hat{f}_k - f_k = S_k^{-1} \sum_i \varepsilon_{i,k}$ and $(\hat{f}_k - f_k)^2 = S_k^{-2} \sum_i \sum_j \varepsilon_{i,k}\varepsilon_{j,k}$.*

Proof. $T_k - f_k S_k = \sum_i \varepsilon_{i,k}$; expand the square of the sum. □

Theorem 5.15 (Estimation variance of the development factor). *Under (M3) and (M2'), with the residual products integrable, on $\{S_k \ne 0\}$, $E[(\hat{f}_k - f_k)^2 \mid \mathcal{D}_k] = \sigma_k^2 / S_k$ (Mack 1993, proof of Theorem 3; Mack 1999, $\mathrm{s.e.}(\hat{f}_k)^2 = \hat{\sigma}_k^2 / S_k$).*

Proof. S_k^{-2} is $\mathcal{D}_k$-measurable and comes out; conditional expectation is linear over the double sum; cross terms vanish by (M2'); diagonal terms give $\sigma_k^2 \sum_i C_{i,k} = \sigma_k^2 S_k$. □

5.5 Unbiasedness of the variance estimator

Lemma 5.16 (Sum of squares around $\hat{f}_k$ via residuals). *With all contributing $C_{i,k} \neq 0$ and $S_k \neq 0$, for any f, $\sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2 = \sum_i \varepsilon_i^2 / C_{i,k} - S_k(\hat{f}_k - f)^2$ where $\varepsilon_i = C_{i,k+1} - f C_{i,k}$.*

Proof. Two instances of Lemma 2.11 (centres $\hat{f}_k$ and f) and $T_k = \hat{f}_k S_k$. □

Definition 5.17. $\hat{\sigma}_k^2$ and the weighted sum of squares $W_k = \sum_i C_{i,k}(F_{i,k} - \hat{f}_k)^2$ as random variables; $\hat{\sigma}_k^2 = W_k/(n-k-2)$.

Theorem 5.18 (Unbiasedness of the variance estimator). *Under (M3) and (M2'), with all contributing $C_{i,k}$ and S_k a.s. nonzero and the stated integrability, $E[W_k \mid \mathcal{D}_k] = (n-k-2)\,\sigma_k^2$ for $k+2 \leq n$, and hence $E[\hat{\sigma}_k^2 \mid \mathcal{D}_k] = \sigma_k^2$ for $k+3 \leq n$.*

Proof. By Lemma 5.16, $W_k = \sum_i \varepsilon_i^2/C_{i,k} - S_k(\hat{f}_k - f_k)^2$. Each $C_{i,k}^{-1}$ is $\mathcal{D}_k$-measurable, so (M3) gives σ_k^2 per term, $n-k-1$ terms in all; S_k is $\mathcal{D}_k$-measurable and Theorem 5.15 gives σ_k^2 for the subtracted term. □

5.6 Process variance and the exact MSEP (Theorem 3)

Lemma 5.19 (Conditional second moment). *Under (M1) and (M3), $E[\varepsilon_{i,k} \mid \mathcal{D}_k] = 0$, $E[C_{i,k+1}^2 \mid \mathcal{D}_k] = f_k^2 C_{i,k}^2 + \sigma_k^2 C_{i,k}$, and for $d \leq k$, $E[C_{i,k+1}^2 \mid \mathcal{D}_d] = f_k^2 E[C_{i,k}^2 \mid \mathcal{D}_d] + \sigma_k^2 E[C_{i,k} \mid \mathcal{D}_d]$.*

Proof. Expand $C_{i,k+1} = f_k C_{i,k} + \varepsilon_{i,k}$; the cross term vanishes; tower property. □

Definition 5.20 (Process variance). $V_0 = 0$, $V_{m+1} = f_{d+m}^2 V_m + \sigma_{d+m}^2\, c \prod_{l<m} f_{d+l}$; in closed form $V_m = \sum_{j<m} \left(\prod_{l=j+1}^{m-1} f_{d+l}\right)^2 \sigma_{d+j}^2\, c \prod_{l<j} f_{d+l}$.

Theorem 5.21 (Process variance along a row). *Under (M1) and (M3), $E[C_{i,d+m}^2 \mid \mathcal{D}_d] - E[C_{i,d+m} \mid \mathcal{D}_d]^2 = V_m$ with $c = C_{i,d}$.*

Proof. Induction on m with Lemma 5.19 and Lemma 5.11. □

Lemma 5.22 (Conditional MSEP decomposition). *For a $\mathcal{D}$-measurable predictor P of Y, $E[(P-Y)^2 \mid \mathcal{D}] = \left(E[Y^2 \mid \mathcal{D}] - E[Y \mid \mathcal{D}]^2\right) + (P - E[Y \mid \mathcal{D}])^2$.*

Proof. Expand the square; P comes out of the conditional expectation. □

Theorem 5.23 (Mack's Theorem 3, exact form). *Let $\mathcal{D}$ be the observed data. Assume $\hat{C}_{i,d+m}$ is $\mathcal{D}$-measurable and that $\mathcal{D}$ adds nothing to $\mathcal{D}_d$ about row i's future (the conditional mean and second moment of $C_{i,d+m}$ given $\mathcal{D}$ agree with those given $\mathcal{D}_d$; this is what independence across accident years supplies). Then under (M1) and (M3)*

$$E\left[(\hat{C}_{i,d+m} - C_{i,d+m})^2 \mid \mathcal{D}\right] = V_m + \left(\hat{C}_{i,d+m} - C_{i,d} \prod_{l<m} f_{d+l}\right)^2.$$

Proof. Lemma 5.22 with $P = \hat{C}_{i,d+m}$, then Theorem 5.21 and Lemma 5.11. □

Definition 5.24 (Mack's estimators of the two terms). $\widehat{V}$ is V_m with $\hat{f}, \hat{\sigma}^2$ plugged in. The estimation-error term $\hat{C}_{i,n-1}^2 \sum_k \hat{\sigma}_k^2/(\hat{f}_k^2 S_k)$ is obtained from $(\hat{C} - M)^2$ by Mack's conditional-resampling step: replace $(\hat{f}_k - f_k)^2$ by σ_k^2/S_k (Theorem 5.15), drop the cross terms (Theorem 5.7), and plug in $\hat{f}, \hat{\sigma}^2$. This is a definition, not a theorem: it is the approximation in Mack (1993), and the exact conditional MSEP above is the object it approximates (Buchwalder, Bühlmann, Merz and Wüthrich 2006 estimate the same object differently).

5.7 From independence across accident years to the $\mathcal{D}_k$ form

Mack states (M1) and (M3) conditioned on one accident year's own history, and assumes separately that the accident years are independent (M2). Every theorem above conditions on $\mathcal{D}_k$ instead. This section proves the passage.

Definition 5.25 (Row σ-algebras)**.** $\sigma(C_{i,0}, \ldots, C_{i,k})$, the history of accident year i up to development year k; $\sigma(C_{i,j} : j)$, the whole of accident year i; and the joins of these over the other accident years $i' \neq i$.

Definition 5.26 ((M2) and the row-conditioned assumptions)**.** (M2) is `RowsIndep`: the σ-algebras of the individual accident years are independent. `RowsGenerateD` says that $\mathcal{D}_k$ is what its name means, the join of the rows' histories up to development year k. (M1row) and (M3row) are Mack's assumptions conditioned on $\sigma(C_{i,0}, \ldots, C_{i,k})$ rather than on $\mathcal{D}_k$.

Lemma 5.27 (Rectangles)**.** *If m_1' and m_2 are independent, F is m_1'-measurable and integrable, $a \in m_1'$ and $b \in m_2$, then $\int_{a \cap b} F = \mu(b) \int_a F$.*

Proof. Apply mathlib's `condExp_indep_eq` to the indicator $1_a F$, which is m_1'-measurable, and integrate the resulting constant over b. $\square$

Theorem 5.28 (Conditioning on an independent enlargement changes nothing)**.** *Let $m_1 \leq m_1'$, let f be m_1'-measurable and integrable, and let m_2 be independent of m_1'. Then $E[f \mid m_1 \vee m_2] = E[f \mid m_1]$ almost surely.*

Proof. The sets $a \cap b$ with $a \in m_1$, $b \in m_2$ form a π-system generating $m_1 \vee m_2$. On such a set both $\int f$ and $\int E[f \mid m_1]$ factor by Lemma 5.27 into $\mu(b)$ times the integral over a, and those agree by the defining property of $E[f \mid m_1]$. The class of sets on which the two set integrals agree is closed under complements and countable disjoint unions, so it is all of $m_1 \vee m_2$; conclude by the uniqueness of the conditional expectation. Mathlib has the case $m_1 = \perp$ only. $\square$

Lemma 5.29 (One row against the others)**.** *Under (M2), accident year i is independent of all the other accident years taken together, and so of their observed histories. Under `RowsGenerateD`, $\mathcal{D}_k$ is the join of row i's history up to k with the other rows' histories up to k.*

Proof. `indep_iSup_of_disjoint` for $\{i\}$ against its complement, then monotonicity of independence; the splitting of $\mathcal{D}_k$ is `iSup_split_single`. $\square$

Theorem 5.30 ((M1) and (M3) in the $\mathcal{D}_k$ form)**.** *Assume (M2), that $\mathcal{D}_k$ is generated by the rows' histories, and the integrability of the entries. Then Mack's row-conditioned (M1row) implies (M1), and (M3row) implies (M3).*

Proof. Split $\mathcal{D}_k$ by Lemma 5.29, apply Theorem 5.28 with m_1 the row's history up to k, m_1' the whole row and m_2 the other rows, and finish with the row-conditioned assumption. $\square$

Theorem 5.31 ((M2') is a consequence, not an assumption)**.** *Under (M2), `RowsGenerateD` and (M1row), the residuals of two different accident years are conditionally uncorrelated given $\mathcal{D}_k$: assumption (M2'), which the variance theorems take as a hypothesis, follows from independence across accident years.*

Proof. Condition on $\mathcal{D}_k$ together with the whole of accident year j: row j's residual comes out of the conditional expectation, and what is left is row i's residual conditioned on a σ-algebra that adds nothing about row i, so it equals $E[\varepsilon_{i,k} \mid \sigma(C_{i,0}, \ldots, C_{i,k})] = 0$ by Theorem 5.28 and (M1row). The tower property brings the identity back to $\mathcal{D}_k$. $\square$

Chapter 6

The variant catalogue: Mack versus conditional resampling

Write $a_k = \hat{\sigma}_k^2/(\hat{f}_k^2 S_k)$ for the relative estimation variance of development factor k along the row. Mack's estimation-error term is $\hat{C}_{i,n-1}^2 \sum_k a_k$; the conditional-resampling term of Buchwalder, Bühlmann, Merz and Wüthrich (2006), which Murphy (1994) also reaches, is $\hat{C}_{i,n-1}^2 \big(\prod_k (1 + a_k) - 1 \big)$.

Definition 6.1. a_k and the conditional-resampling estimator $\hat{C}_{i,n-1}^2 (\prod_k (1 + a_k) - 1)$.

Lemma 6.2 (Product minus sum)**.** *For $a_k \geq 0$: $1 + \sum a_k \leq \prod(1 + a_k) \leq \exp(\sum a_k)$; for two terms $(1 + a)(1 + b) - 1 - (a + b) = ab$; and $\prod(1 + a_k) - 1 - \sum a_k = 0$ when at most one a_k is nonzero.*

Proof. Induction over the index set for the Weierstrass inequality; $1 + x \leq e^x$ termwise for the exponential bound. □

Theorem 6.3 (Catalogue row: Mack is the first-order part of BBMW)**.** *With $a_k \geq 0$ along the row: (i) Mack $\leq$ BBMW; (ii) the difference is exactly $\hat{C}_{i,n-1}^2 \big(\prod_k (1 + a_k) - 1 - \sum_k a_k \big)$; (iii) the two coincide for the second-latest accident year, which has a single development factor; (iv) the difference is at most $\hat{C}_{i,n-1}^2 \big(e^{\sum a_k} - 1 - \sum a_k \big)$, second order in the relative estimation variances.*

Proof. Lemma 6.2 scaled by $\hat{C}_{i,n-1}^2$. □

Theorem 6.4 (Strict counterexample)**.** *There is a run-off triangle (four accident years, four development years, with unequal individual factors in the first two columns) and an accident year on which Mack's estimation-error term is strictly smaller than the conditional-resampling term: the two are different estimators, not two presentations of one. On the exhibited triangle $a_0 = 1/297$, $a_1 = 1/2809$, and the remainder is $\hat{C}^2 a_0 a_1 > 0$. Every number is checked by `norm_num` from the definitions.*

Proof. Evaluate the definitions on the triangle; apply Theorem 6.3(ii). □

6.1 The total reserve: Mack's Corollary

Definition 6.5 (Total-reserve estimators)**.** Mack's Corollary: $\widehat{\mathrm{msep}}(\hat{R}) = \sum_i \widehat{\mathrm{msep}}_i + \sum_i \hat{C}_i \big(\sum_{j>i} \hat{C}_j \big)\, 2 \sum_{k \in \mathrm{row}\, i} a_k$, the cross terms running over the older year's remaining factors. The conditional-resampling aggregate replaces $2 \sum_k a_k$ by $2(\prod_k (1 + a_k) - 1)$ (Buchwalder et al. 2006, Section 4.3).

Theorem 6.6 (Aggregated catalogue row)**.** *The aggregated estimation-error terms differ by* $\sum_i \big(\hat{C}_i^2 + 2\hat{C}_i \sum_{j>i} \hat{C}_j \big) \big(\prod_{k \in \mathrm{row}\, i} (1 + a_k) - 1 - \sum_{k \in \mathrm{row}\, i} a_k \big)$*, which is nonnegative when the ultimates and the* a_k *are: Mack's total is at most the resampling total, by the same row-wise second-order remainder as in the single-year case.*

Proof. Termwise algebra and Lemma 6.2. □

Chapter 7

Solvency II: the one-year view

7.1 The one-year claims development result

Solvency II prices the uncertainty of the next twelve months, not of the run-off to ultimate. Merz and Wüthrich (2008) call the twelve-month move of the predicted ultimate the claims development result and separate the true one from the observable one. The filtration here is the one carried by `RandomTriangle`: $\mathcal{D}_k$ is everything observed up to development year k, for every accident year, the filtration in which (M1) is assumed in this development. Merz and Wüthrich index information by calendar year instead; the martingale property below is the same tower property in either filtration, and the closed form is stated in the filtration (M1) is assumed in.

Definition 7.1 (True one-year claims development result)**.** For accident year i and development year k,

$$\mathrm{CDR}_i(k) = E[C_{i,n-1} \mid \mathcal{D}_k] - E[C_{i,n-1} \mid \mathcal{D}_{k+1}],$$

the change in the predicted ultimate claim when the information grows by one year.

Theorem 7.2 (Merz-Wüthrich 2008: the true CDR has conditional mean zero)**.** *On a finite measure space, $E[\mathrm{CDR}_i(k) \mid \mathcal{D}_k] = 0$ almost surely, for every accident year i and every k: the reserve set at time k is unbiased for the reserve that will be set at time $k+1$ together with the payments made in between.*

Proof. Linearity of the conditional expectation, then the tower property twice: the inner $\mathcal{D}_k$-conditional expectation is idempotent, and $E[E[C_{i,n-1} \mid \mathcal{D}_{k+1}] \mid \mathcal{D}_k] = E[C_{i,n-1} \mid \mathcal{D}_k]$ because $\mathcal{D}_k \subseteq \mathcal{D}_{k+1}$. The two terms cancel. The only hypotheses are that $\mathcal{D}$ is a filtration inside the ambient σ-algebra and that the measure is finite; no integrability assumption on $C_{i,n-1}$ is needed for the identity as stated, since both conditional expectations are 0 off the integrable case. □

Lemma 7.3 (Iterating (M1) from an arbitrary development year)**.** *Under (M1), $E[C_{i,d+m} \mid \mathcal{D}_d] = C_{i,d} \prod_{j=d}^{d+m-1} f_j$ for every starting year d, hence $E[C_{i,n-1} \mid \mathcal{D}_k] = C_{i,k} \prod_{j=k}^{n-2} f_j$ for $k \leq n-1$.*

Proof. The induction of Lemma 5.11 with the starting year free rather than pinned to the latest observed year $n-1-i$ of the row; the base case is that $C_{i,d}$ is $\mathcal{D}_d$-measurable. □

Theorem 7.4 (Merz-Wüthrich 2008: the true CDR as a one-step residual)**.** *Under (M1), for* $k < n-1$ *and integrable claims,*

$$\mathrm{CDR}_i(k) = \Big(\prod_{j=k+1}^{n-2} f_j\Big)\,\big(f_k C_{i,k} - C_{i,k+1}\big)$$

almost surely: everything the next year of experience does to the predicted ultimate is carried by the single deviation of the new observation from its (M1) prediction.

Proof. Both predicted ultimates are given by Lemma 7.3; splitting $\prod_{j=k}^{n-2} f_j = f_k \prod_{j=k+1}^{n-2} f_j$ at the bottom index leaves the common factor $\prod_{j>k} f_j$ times $f_k C_{i,k} - C_{i,k+1}$. □

Definition 7.5 (Observable one-year claims development result)**.** The difference of the two chain-ladder predictions an actuary computes: the ultimate estimated from the triangle with n diagonals minus the ultimate estimated a year later, with every development factor re-estimated, from the triangle with $n+1$ diagonals, $\widehat{\mathrm{CDR}}_i = \hat{C}^{(n)}_{i,n-1} - \hat{C}^{(n+1)}_{i,n-1}$, together with its value along a random triangle outcome by outcome.

Lemma 7.6 (The displayed form)**.** $\widehat{\mathrm{CDR}}_i = \hat{R}^{(n)}_i - \big((C_{i,n-i} - C_{i,n-1-i}) + \hat{R}^{(n+1)}_i\big)$*: opening reserve minus the payments of the year and the closing reserve, the form Merz and Wüthrich display. Algebra on the definitions.*

Proof. Expand both reserves; the two claim entries cancel. □

Nothing is proved here about the distribution of the observable CDR. The conditional mean squared error of prediction of $\widehat{\mathrm{CDR}}_i$, which is what a Solvency II one-year reserve risk figure reports, rests in Merz and Wüthrich on an approximation of the same kind as Mack's: the estimated development factors are treated as resampled ones and terms of second order in the residuals are dropped. That step is not formalized, so the true-CDR statements above are exact and the observable-CDR statements of the paper are not yet available here.

Chapter 8

Röhr 2016: the error-propagation form

Röhr, *Chain ladder and error propagation*, ASTIN Bulletin 46 (2016) 293–330, derives the chain-ladder prediction error from the error-propagation formula. The chain-ladder ultimate $\hat{C}_{i,n-1} = C_{i,d} \prod_{k=d}^{n-2} \hat{f}_k$, $d = n-1-i$, is a product, so a first-order expansion in the development factors turns the relative squared error of the product into the *sum* of the relative squared errors of the factors. The paper's classical-case display is

$$\frac{\widehat{\mathrm{msep}}(\hat{R}_i)}{\hat{C}_{i,n-1}^2} = \sum_k \hat{u}_k^2,$$

where $\hat{u}_k^2$ is the relative uncertainty attached to development step k, and the derivation splits it into a process part and a parameter part,

$$\hat{u}_k^2 = \frac{\hat{\sigma}_k^2}{\hat{f}_k^2\,\hat{C}_{i,k}} + \frac{\hat{\sigma}_k^2}{\hat{f}_k^2\,S_k},$$

the first from the conditional variance $\sigma_k^2 C_{i,k}$ of the step itself, the second from the estimation variance σ_k^2/S_k of $\hat{f}_k$.

Source note. The full text of Röhr (2016) is paywalled and was not available for this formalization. What is formalized is the classical-case (ultimate run-off, single accident year) display quoted above, taken from the paper's own abstract — "in the classical case treated by Mack (1993) the mean squared prediction error divided by the squared estimated ultimate loss can be written as $\sum_j \hat{u}_j^2$, where $\hat{u}_j$ measures the (relative) uncertainty around the j-th development factor and the proportion of the estimated ultimate loss that it affects", with "a split into process error and parameter error" — together with the two parts in the form above. The paper's claims-development-result formulas between two arbitrary future horizons are not formalized here, and neither is its own aggregation over accident years.

8.1 Definitions

Definition 8.1 (Röhr's relative uncertainties). $\hat{u}_{k,\mathrm{proc}}^2 = \hat{\sigma}_k^2/(\hat{f}_k^2 \hat{C}_{i,k})$, $\hat{u}_{k,\mathrm{par}}^2 = \hat{\sigma}_k^2/(\hat{f}_k^2 S_k)$, and $\hat{u}_k^2 = \hat{u}_{k,\mathrm{proc}}^2 + \hat{u}_{k,\mathrm{par}}^2$.

Definition 8.2 (Röhr's linearized prediction error)**.** The process term $\hat{C}_{i,n-1}^2 \sum_{k=d}^{n-2} \hat{u}_{k,\mathrm{proc}}^2$, the parameter term $\hat{C}_{i,n-1}^2 \sum_{k=d}^{n-2} \hat{u}_{k,\mathrm{par}}^2$, and their sum $\mathrm{rohrMsep}_i = \hat{C}_{i,n-1}^2 \sum_{k=d}^{n-2} \hat{u}_k^2$.

8.2 The split, and the two Mack plug-ins

Lemma 8.3 (Process plus parameter)**.** $\mathrm{rohrMsep}_i$ *is the process term plus the parameter term.*

Proof. Termwise, by construction. ☐

Lemma 8.4 (Parameter term is Mack's estimation term)**.** $\hat{u}_{k,\mathrm{par}}^2 = a_k$*, so Röhr's parameter term is Mack's estimation-error term* $\hat{C}_{i,n-1}^2 \sum_k a_k$ *exactly.*

Proof. $\hat{\sigma}_k^2/(\hat{f}_k^2 S_k) = (\hat{\sigma}_k^2/\hat{f}_k^2)/S_k$. ☐

Lemma 8.5 (Splitting the projection)**.** *For* $n-1-i \le k \le m$*,* $\hat{C}_{i,m} = \hat{C}_{i,k} \prod_{l=k}^{m-1} \hat{f}_l$.

Proof. Split the product of the definition at k. ☐

Theorem 8.6 (Process term is Mack's process variance)**.** *Let* $i \le n-1$ *and suppose* $\hat{f}_k \neq 0$ *and* $\hat{C}_{i,k} \neq 0$ *for every* $k = d, \dots, n-2$*. Then Röhr's process term equals* procVar*, Mack's process variance recursion* $V_{m+1} = \hat{f}_{d+m}^2 V_m + \hat{\sigma}_{d+m}^2 \hat{C}_{i,d+m}$*,* $V_0 = 0$*, run for* i *steps with* $\hat{f}, \hat{\sigma}^2$ *plugged in.*

Proof. Write $\hat{C}_{i,n-1} = \hat{C}_{i,k}\, \hat{f}_k \prod_{l=k+1}^{n-2} \hat{f}_l$ by Lemma 8.5. Then $\hat{C}_{i,n-1}^2 \hat{\sigma}_k^2/(\hat{f}_k^2 \hat{C}_{i,k}) = \big(\prod_{l=k+1}^{n-2} \hat{f}_l\big)^2 \hat{\sigma}_k^2 \hat{C}_{i,k}$, which is the k-th summand of the closed form of procVar; reindex $k = d+j$, $j = 0, \dots, i-1$. ☐

8.3 Catalogue row 3

Theorem 8.7 (Röhr = Mack 1993)**.** $\mathrm{rohrMsep}_i = \widehat{\mathrm{msep}}(\hat{R}_i)$*, with no side condition: the linearized error-propagation form and Mack's 1993 closed form are equal term by term. This catalogue row therefore has zero correction term, in contrast with Theorem 6.3. Dividing by* $\hat{C}_{i,n-1}^2 \neq 0$ *recovers Röhr's display* $\widehat{\mathrm{msep}}(\hat{R}_i)/\hat{C}_{i,n-1}^2 = \sum_k \hat{u}_k^2$.

Proof. $\hat{\sigma}_k^2/(\hat{f}_k^2 \hat{C}_{i,k}) + \hat{\sigma}_k^2/(\hat{f}_k^2 S_k) = (\hat{\sigma}_k^2/\hat{f}_k^2)(1/\hat{C}_{i,k} + 1/S_k)$ termwise. ☐

Corollary 8.8 (Mack's closed form is process plus estimation)**.** *Under the hypotheses of Theorem 8.6,* $\widehat{\mathrm{msep}}(\hat{R}_i) = \mathrm{mackProcess}_i + \mathrm{mackEstimation}_i$*: reading Mack's formula through Röhr's split identifies its two halves with the two plug-in estimators of the stochastic layer.*

Proof. Chain Theorem 8.7, Lemma 8.3, Theorem 8.6 and Lemma 8.4. ☐

Theorem 8.9 (What the linearization costs)**.** *Röhr's parameter term is the first-order part of the conditional-resampling term: their difference is exactly* $\hat{C}_{i,n-1}^2 \big(\prod_k (1+a_k) - 1 - \sum_k a_k\big)$*, and with* $a_k \ge 0$ *along the row Röhr's term is the smaller one. Linearizing in the estimated factors is precisely the step that replaces the product by the sum.*

Proof. Lemma 8.4 and Theorem 6.3. ☐

8.4 The total reserve

Definition 8.10 (Aggregate)**.** Röhr's single-year terms carried into Mack's Corollary, with Mack's cross terms: $\sum_i \mathrm{rohrMsep}_i + \sum_i \mathrm{mackCross}_i$. The cross terms are Mack's; the paper's own aggregation over accident years could not be checked against the source, so this is recorded as a definition rather than attributed.

Theorem 8.11. *That aggregate is Mack's total-reserve estimator $\widehat{\mathrm{msep}}(\hat{R})$.*

Proof. Theorem 8.7 row by row. □

8.5 The total reserve: the exact conditional MSEP

Mack's Corollary estimates the mean squared error of prediction of the total reserve $\hat{R} = \sum_i \hat{R}_i$. The estimator is Definition 6.5; the object it estimates is the conditional MSEP of the sum, and this section proves what that object is.

Write $\hat{C}_i = \hat{C}_{i,n-1}$ for the chain-ladder ultimate of accident year i, $C_i = C_{i,n-1}$ for the true ultimate, $\mathcal{D}$ for the σ-algebra of the observed data (the one Theorem 5.23 conditions on) and $M_i = E[C_i \mid \mathcal{D}]$.

Definition 8.12 (Cross-term condition)**.** For a finite set s of accident years and targets Y_i,

$$E\big[(Y_i - E[Y_i \mid \mathcal{D}])(Y_j - E[Y_j \mid \mathcal{D}]) \bigm| \mathcal{D}\big] = 0 \qquad \text{for } i \neq j \text{ in } s.$$

This is what independence across accident years supplies, in the same way that the $\mathcal{D}_k$-conditioned form of (M1) (Definition 5.2) is what it supplies for a single row. Like (M1), (M3) and the "no extra information" hypotheses of Theorem 5.23, it is carried as a hypothesis here; the passage from independence of the rows to it is Theorem 13.6.

Lemma 8.13 (Conditional variance as a centred second moment)**.** *For $Y \in L^2$, $E[(E[Y \mid \mathcal{D}] - Y)^2 \mid \mathcal{D}] = E[Y^2 \mid \mathcal{D}] - E[Y \mid \mathcal{D}]^2$.*

Proof. Lemma 5.22 with the predictor taken to be $E[Y \mid \mathcal{D}]$ itself, so that the estimation-error term is zero. Square integrability supplies every product integrability side condition by Hölder. □

Lemma 8.14 (Conditional variances add)**.** *If $E[Z_i Z_j \mid \mathcal{D}] = 0$ for $i \neq j$ in s and each $Z_i \in L^2$, then $E[(\sum_i Z_i)^2 \mid \mathcal{D}] = \sum_i E[Z_i^2 \mid \mathcal{D}]$.*

Proof. Expand the square into the double sum $\sum_{i,j} Z_i Z_j$, take conditional expectations term by term, and drop the off-diagonal terms. □

Theorem 8.15 (Conditional MSEP of a sum)**.** *For $\mathcal{D}$-measurable predictors P_i of square-integrable targets Y_i whose centred versions satisfy Definition 8.12,*

$$E\Big[\Big(\sum_i P_i - \sum_i Y_i\Big)^2 \Bigm| \mathcal{D}\Big] = \sum_i \big(E[Y_i^2 \mid \mathcal{D}] - E[Y_i \mid \mathcal{D}]^2\big) + \Big(\sum_i \big(P_i - E[Y_i \mid \mathcal{D}]\big)\Big)^2.$$

The conditional variances add; the estimation errors add before *they are squared.*

Proof. Lemma 5.22 applied to the totals, then Lemma 8.13 to rewrite the conditional variance of the total as a centred second moment, then Lemma 8.14 to split it, then Lemma 8.13 again on each accident year. □

Lemma 8.16 (Theorem 1 and the process variance at the ultimate)**.** *Under (M1), (M3) and the hypotheses of Theorem 5.23 that $\mathcal{D}$ carries no information about row i's future beyond $\mathcal{D}_{n-1-i}$,*

$$M_i = C_{i,n-1-i} \prod_{l=n-1-i}^{n-2} f_l, \qquad E[C_i^2 \mid \mathcal{D}] - M_i^2 = \operatorname{procVar}\big(C_{i,n-1-i}, f, \sigma^2, n-1-i\big)_i.$$

Proof. Specialize Lemma 5.11 and Theorem 5.21 to $m = i$, where $(n-1-i)+i = n-1$, and transport them from $\mathcal{D}_{n-1-i}$ to $\mathcal{D}$. □

Theorem 8.17 (Mack's Corollary, exact form)**.** *Under (M1), (M3), square integrability of the claims and of the chain-ladder predictions, the "no extra information" hypotheses of Theorem 5.23 for each accident year, and Definition 8.12 for the true ultimates,*

$$E\Big[\Big(\sum_i \hat{C}_i - \sum_i C_i\Big)^2 \;\Big|\; \mathcal{D}\Big] = \sum_i \operatorname{procVar}_i + \Big(\sum_i (\hat{C}_i - M_i)\Big)^2.$$

The process variances add; the estimation errors add before being squared, and expanding that square is the exact origin of the cross terms of Definition 6.5.

Proof. Theorem 8.15 with $P_i = \hat{C}_i$ and $Y_i = C_i$, then Lemma 8.16 term by term. □

Corollary 8.18 (The total MSEP dominates the process variance)**.** $\sum_i \operatorname{procVar}_i \le E[(\sum_i \hat{C}_i - \sum_i C_i)^2 \mid \mathcal{D}]$ *almost everywhere, whatever the chain-ladder predictions happen to be.*

Proof. The estimation-error term of Theorem 8.17 is a square. □

Theorem 8.19 (The plug-in has the same two-part shape)**.** *On a row whose development factors $\hat{f}_k$ and projections $\hat{C}_{i,k}$ do not vanish, the $1/\hat{C}_{i,k}$ half of Mack's formula is exactly* procVar *with $\hat{f}, \hat{\sigma}^2$ substituted and the $1/S_k$ half is exactly the plug-in estimation error:* $\widehat{\mathrm{msep}}_i = \mathrm{mackProcess}_i + \mathrm{mackEstimation}_i$, *hence* $\widehat{\mathrm{msep}}(\hat{R}) = \sum_i \mathrm{mackProcess}_i + \mathrm{mackTotalEstimation}$. *This is the plug-in counterpart of Theorem 8.17: process variances added, and everything else, single-year terms and cross terms alike, estimating the squared summed estimation error.*

Proof. Reindex the row sum over $\mathrm{Ico}(n-1-i, n-1)$ by $j < i$, split the product $\prod_{l=n-1-i}^{n-2} \hat{f}_l$ at the factor $\hat{f}_{n-1-i+j}$, and cancel; then sum over accident years. □

Theorem 8.20 (The cross-term condition is not vacuous)**.** *On the nondegenerate model of Theorem 15.1 the three true ultimates are genuinely random and their centred versions are conditionally uncorrelated given the trivial σ-algebra: Definition 8.12 holds for $s = \{0, 1, 2\}$ without holding for want of randomness.*

Proof. The unconditional mean of each ultimate is cf^2; the centred ultimates are proportional to the shocks ξ_i, which are orthogonal. □

Chapter 9

Mack 1999: general weights and exponent

[2] generalizes the development factor of [1] to

$$\hat{f}_k = \frac{\sum_i w_{ik} C_{ik}^{\alpha} F_{ik}}{\sum_i w_{ik} C_{ik}^{\alpha}}, \qquad F_{ik} = \frac{C_{i,k+1}}{C_{ik}},$$

with weights $w_{ik} \in [0,1]$ and an exponent α. The three cases the paper names are $\alpha = 1$ with unit weights (the 1993 estimator), $\alpha = 0$ (the simple average of the individual factors) and $\alpha = 2$ (the ordinary regression of $C_{i,k+1}$ on C_{ik} through the origin). The variance assumption becomes $\mathrm{Var}(F_{ik} \mid \cdot) = \sigma_k^2/(w_{ik} C_{ik}^{\alpha})$ and the variance estimator

$$\hat{\sigma}_k^2 = \frac{1}{n-k-2} \sum_i w_{ik} C_{ik}^{\alpha} \, (F_{ik} - \hat{f}_k)^2.$$

Two displays of that paper are formalized here. The first is its equation $(*)$,

$$\text{s.e.}(\hat{C}_{in})^2 = \hat{C}_{in}^2 \sum_k \frac{\hat{\sigma}_k^2}{\hat{f}_k^2} \Big(\frac{1}{w_{ik} \hat{C}_{ik}^{\alpha}} + \frac{1}{\sum_j w_{jk} C_{jk}^{\alpha}} \Big),$$

the second is the recursion displayed immediately below it,

$$\text{s.e.}(\hat{C}_{i,k+1})^2 = \hat{C}_{ik}^2 \big(\text{s.e.}(F_{ik})^2 + \text{s.e.}(\hat{f}_k)^2\big) + \text{s.e.}(\hat{C}_{ik})^2 \hat{f}_k^2, \qquad \text{s.e.}(F_{ik})^2 = \frac{\hat{\sigma}_k^2}{w_{ik} \hat{C}_{ik}^{\alpha}}, \quad \text{s.e.}(\hat{f}_k)^2 = \frac{\hat{\sigma}_k^2}{\sum_j w_{jk} C_{jk}^{\alpha}}.$$

Theorem 3.5 proves the two agree for $\alpha = 1$ and unit weights; the theorem below carries the weights and the exponent through, and the unit case recovers the earlier statement. Mack's restriction $w_{ik} \in [0,1]$ and $\alpha \in \{0,1,2\}$ is a modelling choice and no identity here uses it, so it is not imposed. Section 3 of the paper, the tail factor, is a convention: the tail factor and the variance parameters attached to it are supplied by the actuary, so they are definitions with no theorem attached.

9.1 Definitions

Definition 9.1 (Weighted column sums). $S_k^{w,\alpha} = \sum_{i=0}^{n-k-2} w_{ik} C_{ik}^{\alpha}$ and $T_k^{w,\alpha} = \sum_{i=0}^{n-k-2} w_{ik} C_{ik}^{\alpha} F_{ik}$; `unitWeights` is $w_{ik} = 1$.

Definition 9.2 (Weighted development factor and variance estimator). $\hat{f}_k = T_k^{w,\alpha}/S_k^{w,\alpha}$ and $\hat{\sigma}_k^2 = \frac{1}{n-k-2}\sum_i w_{ik}C_{ik}^{\alpha}(F_{ik}-\hat{f}_k)^2$. As in the 1993 case, division by zero is zero, and the value at $k = n-2$ is an extrapolation convention rather than an estimator.

Definition 9.3 (Weighted projection and ultimate). $\hat{C}_{ik} = C_{i,n-1-i}\prod_{j=n-1-i}^{k-1}\hat{f}_j$ and $\hat{C}_{i,n-1}$, both built from the weighted factors.

Definition 9.4 (Mack 1999, equation $(*)$).

$$\mathrm{msepW}_i = \hat{C}_{i,n-1}^2 \sum_{k=n-1-i}^{n-2} \frac{\hat{\sigma}_k^2}{\hat{f}_k^2}\Big(\frac{1}{w_{ik}\hat{C}_{ik}^{\alpha}} + \frac{1}{S_k^{w,\alpha}}\Big).$$

Definition 9.5 (Mack 1999, the recursion below $(*)$). m steps of $V_{m+1} = \hat{C}_{ik}^2\big(\hat{\sigma}_k^2/(w_{ik}\hat{C}_{ik}^{\alpha}) + \hat{\sigma}_k^2/S_k^{w,\alpha}\big) + V_m\hat{f}_k^2$ with $k = n-1-i+m$, started at $V_0 = 0$ on the latest observed diagonal.

Definition 9.6 (Tail factor, Section 3). The ultimate extended by a tail factor, $\hat{C}_{i,\infty} = \hat{C}_{i,n-1}f_{\mathrm{tail}}$, and one further step of the recursion with actuary-supplied $f_{\mathrm{tail}}, \sigma_{\mathrm{tail}}^2, S_{\mathrm{tail}}$. The paper gives no estimator for these, only the arithmetic of carrying them through; they are conventions and no theorem here uses them.

9.2 Deterministic identities

Lemma 9.7 (Weighted-average form). $\hat{f}_k = \sum_i\big(w_{ik}C_{ik}^{\alpha}/S_k^{w,\alpha}\big)F_{ik}$*: the generalized factor is the* $w_{ik}C_{ik}^{\alpha}$*-weighted mean of the individual factors. This is Lemma 2.9 with the weights carried through; because the numerator is written with the individual factors, no nonvanishing hypothesis is needed.*

Proof. Divide the numerator termwise by $S_k^{w,\alpha}$. □

Lemma 9.8 (Weighted sum of squares). *For any centre* f, $\sum_i w_{ik}C_{ik}^{\alpha}(F_{ik}-f)^2 = \sum_i w_{ik}C_{ik}^{\alpha}F_{ik}^2 - 2fT_k^{w,\alpha} + f^2S_k^{w,\alpha}$*; at* $f = \hat{f}_k$*, and provided* $S_k^{w,\alpha} \neq 0$*, the cross term collapses to* $\sum_i w_{ik}C_{ik}^{\alpha}F_{ik}^2 - S_k^{w,\alpha}\hat{f}_k^2$*. These generalize Lemmas 2.11 and 2.12, the identity behind the* $n-k-2$ *degrees of freedom.*

Proof. Expand termwise; substitute $\hat{f}_k = T_k^{w,\alpha}/S_k^{w,\alpha}$. □

Lemma 9.9 (Weighted projection identities). $\hat{C}_{i,k+1} = \hat{C}_{ik}\hat{f}_k$ *for* $k \geq n-1-i$, $\hat{C}_{i,n-1-i} = C_{i,n-1-i}$*, and* $(*)$ *is the ultimate squared times the sum of its summands.*

Proof. Split the product at the top; the empty product is one. □

9.3 The recursion equals $(*)$

Lemma 9.10 (One recursion step). *For reals* A, B, P, S, f, X *with* $f \neq 0$,

$$A^2\Big(\frac{B}{P} + \frac{B}{S}\Big) + A^2Xf^2 = (Af)^2\Big(X + \frac{B}{f^2}\Big(\frac{1}{P} + \frac{1}{S}\Big)\Big).$$

With $A = \hat{C}_{ik}$, $f = \hat{f}_k$, $B = \hat{\sigma}_k^2$, $P = w_{ik}\hat{C}_{ik}^{\alpha}$ *and* $S = S_k^{w,\alpha}$*: one step of the recursion adds exactly the* k*-th summand of* $(*)$*, scaled by the next projection. Only* $\hat{f}_k \neq 0$ *is used.*

Proof. Clear the f^2 and expand. $\square$

Lemma 9.11 (Truncated form)**.** *If $\hat{f}_k \neq 0$ for $k = n-1-i, \ldots, n-2-i+m$, then m steps of the recursion equal $\hat{C}^2_{i,n-1-i+m} \sum_{k=n-1-i}^{n-2-i+m}$ of the summands of $(*)$.*

Proof. Induction on m, using Lemma 9.10 at each step. $\square$

Theorem 9.12 (Mack 1999: the recursion is $(*)$, for general w and α)**.** *Let $i \leq n-1$ and suppose $\hat{f}_k \neq 0$ for $k = n-1-i, \ldots, n-2$. Then i steps of the recursion displayed below Mack's $(*)$ return $(*)$ itself, for every weight function and every exponent.*

Proof. Lemma 9.11 at $m = i$, where $n-1-i+i = n-1$. $\square$

9.4 The 1993 case

Lemma 9.13 (Unit weights, $\alpha = 1$)**.** *At $\alpha = 1$ with $w \equiv 1$, and provided the contributing entries C_{ik} do not vanish, $S_k^{w,\alpha} = S_k$, $T_k^{w,\alpha} = T_k$, $\hat{f}_k$ is the 1993 factor, $\hat{\sigma}_k^2$ is Mack's 1993 variance estimator, and the projection, the ultimate and the summands of $(*)$ are the 1993 ones.*

Proof. Termwise; $C_{ik}F_{ik} = C_{i,k+1}$ needs $C_{ik} \neq 0$. $\square$

Theorem 9.14 (Specialization to Mack 1993)**.** *At $\alpha = 1$ with unit weights, $(*)$ is $\widehat{\mathrm{msep}}(\hat{R}_i)$ and the generalized recursion is the recursion of Theorem 3.5; consequently i steps of it return Mack's 1993 estimator, which is the statement of Theorem 3.5.*

Proof. Rewrite with Lemma 9.13 and apply Theorem 9.12. $\square$

9.5 Conditional unbiasedness of the weighted estimator

Definition 9.15 (The weighted estimator as a random variable)**.** $S_k^{w,\alpha}$ and $\hat{f}_k$ evaluated at the random triangle, with weights $w_{ik} : \Omega \to \mathbb{R}$, together with the multiplier $g_{ik} = w_{ik}C_{ik}^{\alpha}/C_{ik}$ of $C_{i,k+1}$ inside $\hat{f}_k$. For $\alpha = 1$ and unit weights $g_{ik} = 1$; writing the multiplier this way keeps $\alpha = 0$, where it is $1/C_{ik}$, in range.

Lemma 9.16 (Measurability and the linear form)**.** *If every w_{ik} is $\mathcal{D}_k$-measurable then so are $S_k^{w,\alpha}$ and g_{ik} (for $\alpha = 2$ this uses only measurability of C_{ik}, since C_{ik}^2 is then measurable), and $\hat{f}_k = (S_k^{w,\alpha})^{-1} \sum_i g_{ik}C_{i,k+1}$: the weighted estimator is a $\mathcal{D}_k$-measurable linear combination of the next column.*

Proof. Sums, products, powers and quotients of measurable functions; the linear form is $w_{ik}C_{ik}^{\alpha}F_{ik} = (w_{ik}C_{ik}^{\alpha}/C_{ik})C_{i,k+1}$. $\square$

Theorem 9.17 (The weighted estimator is conditionally unbiased)**.** *Under (M1) in the $\mathcal{D}_k$ form, with $\mathcal{D}_k$-measurable weights, on the event that the weighted column sum is nonzero and where the contributing entries C_{ik} do not vanish, $E[\hat{f}_k \mid \mathcal{D}_k] = f_k$ for every exponent α. This is Theorem 5.5 with the weights and the exponent carried through.*

Proof. By Lemma 9.16 the factor $(S_k^{w,\alpha})^{-1}$ is $\mathcal{D}_k$-measurable and comes out of the conditional expectation, and so does each g_{ik}. (M1) turns $E[C_{i,k+1} \mid \mathcal{D}_k]$ into f_kC_{ik}, and $g_{ik}C_{ik} = w_{ik}C_{ik}^{\alpha}$ when $C_{ik} \neq 0$, so the sum is $f_kS_k^{w,\alpha}$. The hypothesis $C_{ik} \neq 0$ is what the individual factors need in any case and cannot be dropped at $\alpha = 0$. $\square$

Chapter 10

The over-dispersed Poisson model

Renshaw and Verrall, *A stochastic model underlying the chain-ladder technique*, British Actuarial Journal 4 (1998) 903–923, put the run-off triangle in incremental form, $X_{i,k} = C_{i,k} - C_{i,k-1}$ with $X_{i,0} = C_{i,0}$, and model the incrementals as an over-dispersed Poisson (ODP) generalized linear model: independent, with mean $m_{i,k}$ and variance $\varphi\, m_{i,k}$, and a log link carrying a row effect and a column effect, so the mean is multiplicative,

$$m_{i,k} = a_i b_k, \qquad \sum_{k=0}^{n-1} b_k = 1.$$

Mack, *A simple parametric model for rating automobile insurance or estimating IBNR claims reserves*, ASTIN Bulletin 21 (1991) 93–109, starts from the same multiplicative mean and estimates it from the *marginal sums*. Both papers arrive at the same system: for each accident year i and each development year k,

$$\sum_{k\,:\,i+k\le n-1} m_{i,k} = \sum_{k\,:\,i+k\le n-1} X_{i,k}, \qquad \sum_{i\,:\,i+k\le n-1} m_{i,k} = \sum_{i\,:\,i+k\le n-1} X_{i,k}.$$

These are the score equations of the quasi-Poisson log-likelihood $\sum(X \log m - m)/\varphi$ in the row and column parameters; the dispersion φ multiplies the whole score and cancels, which is why the ODP fit is the Poisson fit and why the estimates do not depend on φ.

The chapter proves the two halves of the classical equivalence. Chain ladder *solves* the system (Theorems 10.10 and 10.12), and a positive multiplicative solution of the system *is* chain ladder (Theorem 10.16).

Source note. Neither paper was available in full text for this formalization. What is formalized is the marginal-sum system as quoted above and as it is stated in the secondary literature (England and Verrall, *Stochastic claims reserving in general insurance*, British Actuarial Journal 8 (2002) 443–518, Section 2.3; Wüthrich and Merz, *Stochastic Claims Reserving Methods in Insurance* (2008), Section 2.3), in the two papers' own terms. The displayed equations of Renshaw and Verrall (Section 3) and of Mack (Section 2) are therefore identified by section and by the display itself, not by an equation number. Nothing here is probabilistic: the ODP assumption enters only through the shape of the score equations, which are algebraic identities in the fitted values.

10.1 Definitions

Definition 10.1 (Incremental claims). $X_{i,k} = C_{i,k} - C_{i,k-1}$ for $k \geq 1$ and $X_{i,0} = C_{i,0}$.

Definition 10.2 (Observed row and column)**.** The cell (i, k) is observed when $i + k \le n - 1$. The observed development years of accident year i are $\mathrm{obsRow}(n, i) = \{0, \ldots, n-1-i\}$ and the observed accident years of development year k are $\mathrm{obsCol}(n, k) = \{0, \ldots, n-1-k\}$. The second one at $k + 1$ is the contributor set of $\hat{f}_k$ (`obsCol_succ`).

Definition 10.3 (Chain-ladder fit on the observed triangle)**.** The fitted cumulative on an observed cell is the latest observed entry of the row divided back by the development factors,

$$\hat{C}_{i,k} = \frac{C_{i,\,n-1-i}}{\prod_{j=k}^{n-2-i} \hat{f}_j},$$

and the fitted incremental is $\hat{X}_{i,k} = \hat{C}_{i,k} - \hat{C}_{i,k-1}$ with $\hat{X}_{i,0} = \hat{C}_{i,0}$. These are the $\hat{C}$ projections of Definition 2.6 read backwards from the latest diagonal: equivalently $\hat{C}_{i,k} = \hat{C}_{i,n-1} / \prod_{j=k}^{n-2} \hat{f}_j$, the chain-ladder ultimate times the cumulative development pattern up to k (`CLcum_eq_ultimate_div`).

Definition 10.4 (Multiplicative model and its pattern)**.** $m_{i,k} = a_i b_k$; the identifiability constraint $\sum_{k<n} b_k = 1$; and the cumulative pattern $B_k = \sum_{j \le k} b_j$.

Definition 10.5 (Score (marginal-sum) equations)**.** The row marginal sums, the column marginal sums, and their conjunction: the fitted values reproduce the observed row and column totals of incrementals on the observed triangle.

Definition 10.6 (Quasi-Poisson log-likelihood)**.** $\sum_k \big(X_{i,k} \log(t\, b_k) - t\, b_k\big)$ as a function of the row parameter t, and $\sum_i \big(X_{i,k} \log(a_i\, t) - a_i\, t\big)$ as a function of the column parameter t (dispersion $\varphi = 1$, which only scales the expression).

10.2 The score equations are score equations

Lemma 10.7 (Row and column scores)**.** *The derivative of the quasi-Poisson log-likelihood in a row parameter is $\big(\sum_k X_{i,k} - \sum_k t\, b_k\big)/t$, and in a column parameter $\big(\sum_i X_{i,k} - \sum_i a_i\, t\big)/t$.*

Proof. Termwise: $\frac{d}{dt}\big(X \log(tb) - tb\big) = X/t - b$, then sum over the observed cells. □

Lemma 10.8 (Stationarity is the marginal-sum equation)**.** *For nonzero parameters, the row score vanishes exactly at the row marginal-sum equation and the column score exactly at the column one. This is what licenses the name "score equations" for Definition 10.5, and it is where the dispersion φ drops out.*

Proof. Lemma 10.7 and $a/b = 0 \iff a = 0$ for $b \ne 0$. □

10.3 Chain ladder solves the score equations

Lemma 10.9 (Telescoping)**.** $\sum_{k \le m} X_{i,k} = C_{i,m}$ *and* $\sum_{k \le m} \hat{X}_{i,k} = \hat{C}_{i,m}$.

Proof. Induction on m. □

Theorem 10.10 (Row totals)**.** *For every accident year i, $\sum_{k \le n-1-i} \hat{X}_{i,k} = C_{i,\,n-1-i} = \sum_{k \le n-1-i} X_{i,k}$: the chain-ladder fit reproduces every observed row total. This is the first marginal-sum family of Mack (1991), Section 2, and the row score equation of Renshaw and Verrall (1998), Section 3. No hypothesis is needed.*

Proof. Both sides telescope to the latest observed entry, by Lemma 10.9 and $\hat{C}_{i,\,n-1-i} = C_{i,\,n-1-i}$ (the empty product). □

Lemma 10.11 (Fitted column sums of cumulatives)**.** *If every development factor $\hat{f}_j$, $j < n-1$, is nonzero, then for every $k \le n-1$*

$$\sum_{i \le n-1-k} \hat{C}_{i,k} = \sum_{i \le n-1-k} C_{i,k},$$

and, one year earlier over the same accident years, $\sum_{i \le n-2-k} \hat{C}_{i,k} = S_k$.

Proof. Downward induction on k. At $k = n-1$ the only observed cell is the corner, where the fit is the observation. For the step, the cells strictly above the diagonal satisfy $\hat{C}_{i,k} = \hat{C}_{i,k+1}/\hat{f}_k$, so their sum is $T_k/\hat{f}_k = S_k$ by the induction hypothesis and the definition $\hat{f}_k = T_k/S_k$; adding the diagonal cell, where the fit is the observation, gives the claim. Nonvanishing of $\hat{f}_k$ forces $T_k \neq 0$ and $S_k \neq 0$, which is the only place a hypothesis is used. □

Theorem 10.12 (Column totals)**.** *If every development factor $\hat{f}_j$, $j < n-1$, is nonzero, then for every development year k*

$$\sum_{i \le n-1-k} \hat{X}_{i,k} = \sum_{i \le n-1-k} X_{i,k} :$$

the chain-ladder fit reproduces every observed column total. This is the second marginal-sum family of Mack (1991), Section 2, and the column score equation of Renshaw and Verrall (1998), Section 3.

Proof. For $k = 0$ the incrementals are the cumulatives and this is Lemma 10.11. For $k \ge 1$ subtract the two displays of Lemma 10.11: the first over the observed part of column k, the second at $k-1$ over the same accident years. □

Theorem 10.13 (Chain ladder solves the ODP score equations)**.** *Under the same hypothesis, the chain-ladder fitted incrementals satisfy the whole system of Definition 10.5.*

Proof. Theorems 10.10 and 10.12. □

10.4 Uniqueness: the marginal-sum solution is chain ladder

Mack (1991), Section 2, states the converse: the marginal-sum estimates of the multiplicative model are the chain-ladder estimates. Positivity of the parameters replaces the informal assumption that the triangle is a real one.

Lemma 10.14 (The pattern ratios are the development factors)**.** *Let $a_i, b_k > 0$ with $\sum_{k<n} b_k = 1$ satisfy the score equations. Then for every $k \le n-1$*

$$\Big(\sum_{i \le n-1-k} a_i\Big) B_k \ \textit{splits as} \sum_{i \le n-1-k} C_{i,k}, \qquad \textit{and} \qquad B_k \hat{f}_k = B_{k+1}\ (k < n-1),$$

so every $\hat{f}_k$ is positive.

Proof. Downward induction on k, mirroring Lemma 10.11. The row equation for accident year i reads $a_i B_{n-1-i} = C_{i,\,n-1-i}$; at $i = 0$, with $B_{n-1} = 1$, it starts the induction. In the step, write $A = \sum_{i \le n-2-k} a_i$. The induction hypothesis gives $AB_{k+1} = T_k$ and the column equation at $k+1$ gives $Ab_{k+1} = T_k - S_k$, so $AB_k = S_k$; adding the diagonal cell, whose row equation is $a_{n-1-k} B_k = C_{n-1-k,k}$, closes the induction. Dividing the two displays, $\hat{f}_k = T_k/S_k = B_{k+1}/B_k$. □

Lemma 10.15 (Fitted cumulatives agree). *Under the hypotheses of Lemma 10.14, $a_i B_k = \hat{C}_{i,k}$ on every observed cell.*

Proof. Telescoping $B_k \prod_{j=k}^{d-1} \hat{f}_j = B_d$ with $d = n - 1 - i$, and the row equation $a_i B_d = C_{i,d}$; the product is nonzero because each factor is. □

Theorem 10.16 (Mack's uniqueness result). *A multiplicative model $m_{i,k} = a_i b_k$ with $a_i, b_k > 0$ and $\sum_{k<n} b_k = 1$ that satisfies the marginal-sum equations coincides with the chain-ladder fit cell by cell on the observed triangle: $a_i b_k = \hat{X}_{i,k}$. With Theorem 10.13 this is the classical equivalence: the over-dispersed Poisson generalized linear model and the chain-ladder method give the same fitted values, hence the same development pattern and the same reserves.*

Proof. Differencing Lemma 10.15 at k and $k - 1$, since $B_k - B_{k-1} = b_k$. □

Chapter 11

A finite independent-row witness

The passage from Mack's row-conditioned assumptions to the observed-data filtration is realized by a concrete model. This section supplies one finite probability space satisfying every hypothesis used by Theorems 5.30 and 5.31. The model is a repository witness, while the assumptions it realizes are Mack's equations (1) and (2) and the filtration B_k used in the proof of Theorem 2 [1].

Definition 11.1 (Eight-outcome triangle with row-generated filtration)**.** There are three independent sign coordinates on eight equally likely outcomes. Claims in accident year i are constant at development year zero and depend only on sign ξ_i thereafter. The filtration at development year k is defined to be the join of the claims in every row through k. The positive-sign event in row i is called `shockEvent i`.

Theorem 11.2 (Independent rows and the generated filtration)**.** *The three row shocks are mutually independent. Every claim in row i is measurable with respect to that row's shock, so the full row sigma-algebras are independent. By construction, the observed-data filtration is exactly the join of the row histories.*

Proof. The eight intersections of the three positive-sign events have the product probabilities 1, 1/2, 1/4 and 1/8 according to their cardinality. Independence passes to the smaller sigma-algebras generated by the claims. The filtration identity is definitional. □

Theorem 11.3 (The independence passage is non-vacuous)**.** *The model satisfies Mack's row-conditioned mean equation (1). The general independence theorems therefore give the corresponding mean equation conditioned on $\mathcal{D}_k$ and the cross-row residual condition (M2'). Two outcomes give different claims in the same accident year, so this is a genuinely random instance of both conclusions.*

Proof. At the first development step, conditional expectation over the trivial row history averages the two signs. Every later claim is the stated development factor times the previous claim and is already measurable in the row history. Apply Theorems 5.30 and 5.31; direct evaluation at outcomes zero and one proves nontriviality. □

Chapter 12

Merz-Wüthrich: the MSEP of the one-year claims development result

Merz and Wüthrich (2008), *Modelling the claims development result for solvency purposes*, Casualty Actuarial Society E-Forum, Fall 2008, pages 542-568, quantify the uncertainty of the one-year move of the predicted ultimate. The previous chapter defined the true claims development result and proved that it has conditional mean zero; this chapter adds its second moment, which is the conditional mean squared error of predicting it by the budget value 0, and records the estimator the paper displays for the observable claims development result.

Index conventions. The paper writes I for the last accident year, $J = I$ for the last development year, and $\mathcal{D}_I$ for the triangle after I diagonals. Here $I = J = n - 1$; $I - i$ is $n - 1 - i$, $C_{I-j,j}$ is $C_{n-1-j,\,j}$, S_j^I is S_j computed with n accident years and S_j^{I+1} is the same column sum computed with $n + 1$, that is, with the next diagonal added. Both are functions of the triangle observed today, because $S_j^{I+1} = S_j^I + C_{I-j,j}$ and $C_{I-j,j}$ sits on the latest observed diagonal.

The step formalized below is the development-year step $\mathcal{D}_k \to \mathcal{D}_{k+1}$ of this library's filtration, not the paper's calendar-year step; the caveat is the one recorded for Theorem 7.4.

12.1 The exact one-year process variance

Definition 12.1 (One-year process variance)**.** For claims c at development year k and ultimate development year N,

$$\mathrm{cdrProcVar}(c, f, \sigma^2, N, k) = \Big(\prod_{j=k+1}^{N-1} f_j \Big)^2 \sigma_k^2\, c.$$

Theorem 12.2 (Merz-Wüthrich 2008, display (2.18))**.** *Under (M1) and (M3), for $i < n$ and $k < n - 1$ and integrable claims,*

$$E\big[\mathrm{CDR}_i(k)^2 \mid \mathcal{D}_k\big] = \Big(\prod_{j=k+1}^{n-2} f_j \Big)^2 \sigma_k^2\, C_{i,k}$$

almost surely. Since the true CDR has conditional mean zero, this is at once its conditional variance and the conditional mean squared error of predicting it by 0, which is the budget value the paper enters in the income statement for prior accident years.

Proof. By Theorem 7.4 the true CDR is $\left(\prod_{j>k} f_j\right)$ times the one-step residual $\varepsilon_{i,k} = C_{i,k+1} - f_k C_{i,k}$, up to sign. The constant comes out of the conditional expectation squared, and $E[\varepsilon_{i,k}^2 \mid \mathcal{D}_k] = \sigma_k^2 C_{i,k}$ is assumption (M3). The conditional-variance form subtracts Theorem 7.2 squared, which is 0. ☐

12.2 The estimator of Merz and Wüthrich

Definitions only. The derivation in the paper's appendix replaces the estimated development factors by resampled ones and drops terms of higher order in the residuals, so none of the objects below is upgraded to an identity.

Lemma 12.3 (The column sum one year later)**.** *For $j < n$, $S_j^{I+1} = S_j^I + C_{I-j,j}$.*

Proof. The index set gains exactly the row $I - j$. ☐

Definition 12.4 (Displays (3.4)-(3.7))**.** With $d = I - i$,

$$\hat{\Psi}_i^I = \frac{\hat{\sigma}_d^2/(\hat{f}_d^I)^2}{C_{i,d}}, \qquad \hat{\Phi}_{i,J}^I = \sum_{j=d+1}^{J-1} \Big(\frac{C_{I-j,j}}{S_j^{I+1}}\Big)^2 \frac{\hat{\sigma}_j^2/(\hat{f}_j^I)^2}{C_{I-j,j}},$$

$$\hat{\Gamma}_{i,J}^I = \hat{\Phi}_{i,J}^I + \hat{\Psi}_i^I, \qquad \hat{\Delta}_{i,J}^I = \frac{\hat{\sigma}_d^2/(\hat{f}_d^I)^2}{S_d^I} + \sum_{j=d+1}^{J-1} \Big(\frac{C_{I-j,j}}{S_j^{I+1}}\Big)^2 \frac{\hat{\sigma}_j^2/(\hat{f}_j^I)^2}{S_j^I}.$$

The later development steps enter the one-year figure only through the re-estimation of their development factors on the next diagonal, which is why each is scaled by $(C_{I-j,j}/S_j^{I+1})^2 \le 1$.

Definition 12.5 (Displays (3.8)-(3.10))**.** $\widehat{\mathrm{Var}}(\mathrm{CDR}_i(I+1) \mid \mathcal{D}_I) = (\hat{C}_{i,J}^I)^2 \hat{\Psi}_i^I$; the prospective estimator $\widehat{\mathrm{msep}}(0) = (\hat{C}_{i,J}^I)^2(\hat{\Gamma}_{i,J}^I + \hat{\Delta}_{i,J}^I)$ split into its process part $(\hat{C}_{i,J}^I)^2\hat{\Gamma}_{i,J}^I$ and its parameter part $(\hat{C}_{i,J}^I)^2\hat{\Delta}_{i,J}^I$; and the retrospective estimator $\widehat{\mathrm{msep}}(\widehat{\mathrm{CDR}}_i(I+1)) = (\hat{C}_{i,J}^I)^2(\hat{\Phi}_{i,J}^I + \hat{\Delta}_{i,J}^I)$.

Lemma 12.6 (Display (3.11))**.** $\widehat{\mathrm{msep}}(0) = \widehat{\mathrm{msep}}(\widehat{\mathrm{CDR}}_i(I+1)) + \widehat{\mathrm{Var}}(\mathrm{CDR}_i(I+1) \mid \mathcal{D}_I)$*, and the second is at most the first when $\hat{\Psi}_i^I \ge 0$.*

Proof. $\hat{\Gamma} = \hat{\Phi} + \hat{\Psi}$. ☐

Theorem 12.7 (Display (3.17))**.** *If the column sums S_j^I and S_j^{I+1} along the row do not vanish, then*

$$\widehat{\mathrm{msep}}(0) = (\hat{C}_{i,J}^I)^2 \Big(\frac{\hat{\sigma}_d^2/(\hat{f}_d^I)^2}{C_{i,d}} + \frac{\hat{\sigma}_d^2/(\hat{f}_d^I)^2}{S_d^I} + \sum_{j=d+1}^{J-1} \frac{C_{I-j,j}}{S_j^{I+1}} \frac{\hat{\sigma}_j^2/(\hat{f}_j^I)^2}{S_j^I} \Big).$$

Read next to Mack's Theorem 3, the one-year figure keeps only the first term of the process variance, keeps the estimation error of the next diagonal in full, and scales every remaining run-off cell by $C_{I-j,j}/S_j^{I+1} \le 1$.

Proof. The two sums combine term by term: with $s = S_j^I$, $c = C_{I-j,j}$ and $s + c = S_j^{I+1}$ by Lemma 12.3, $(c/(s+c))^2 a/c + (c/(s+c))^2 a/s = (c/(s+c))\, a/s$. ☐

12.3 The bridge to the exact statement

Theorem 12.8 (The first development step is the plug-in). *If the row is not fully developed and $\hat{f}^I_d \neq 0$, $C_{i,d} \neq 0$, then $(\hat{C}^I_{i,J})^2\hat{\Psi}^I_i$ is exactly* $\mathrm{cdrProcVar}(C_{i,d}, \hat{f}, \hat{\sigma}^2, n-1, d)$*: display (3.8) is the plug-in of Theorem 12.2 for the step the next calendar year observes. The process part of display (3.9) is that plug-in plus the correction $(\hat{C}^I_{i,J})^2\hat{\Phi}^I_{i,J}$ for the later steps.*

Proof. Split the ultimate at its bottom factor, $\hat{C}^I_{i,J} = C_{i,d}\,\hat{f}^I_d \prod_{j>d} \hat{f}^I_j$; the squares of $C_{i,d}$ and $\hat{f}^I_d$ cancel against the denominators of $\hat{\Psi}^I_i$. □

Theorem 12.9 (The one-year process variances telescope). *Summing the exact one-year process variance of Theorem 12.2 over every remaining development step of the row, with $C_{i,k}$ replaced by its (M1) prediction $C_{i,d}\prod_{l=d}^{k-1} f_l$, gives Mack's multi-year process variance V_m exactly, with no remainder. The stochastic form performs the replacement by a conditional expectation given $\mathcal{D}_d$ instead of by hand.*

Proof. Term by term this is the closed form of V_m (Definition 5.20): its j-th summand is the one-year process variance of step $d+j$. For the stochastic form, each summand is a constant times $C_{i,d+j}$, and Lemma 7.3 evaluates its conditional expectation. □

12.4 Aggregation over accident years

Accident year 0 is fully developed, so the sums run over $i = 1, \ldots, n-1$ and the cross terms over the pairs $k > i > 0$, the older year i carrying the shared development factors. Definitions only.

Definition 12.10 (Displays (3.12)-(3.15)).

$$\hat{\Lambda}^I_{i,J} = \frac{C_{i,d}}{S^{I+1}_d}\,\frac{\hat{\sigma}^2_d/(\hat{f}^I_d)^2}{S^I_d} + \sum_{j=d+1}^{J-1}\left(\frac{C_{I-j,j}}{S^{I+1}_j}\right)^2 \frac{\hat{\sigma}^2_j/(\hat{f}^I_j)^2}{S^I_j}, \qquad \hat{\Xi}^I_{i,J} = \hat{\Phi}^I_{i,J} + \frac{\hat{\sigma}^2_d/(\hat{f}^I_d)^2}{S^{I+1}_d},$$

and the two aggregates $\sum_i \widehat{\mathrm{msep}}_i + 2\sum_{k>i>0} \hat{C}^I_{i,J}\hat{C}^I_{k,J}(\cdot_i + \hat{\Lambda}^I_{i,J})$ with $\cdot_i = \hat{\Phi}^I_{i,J}$ for the retrospective view (3.12) and $\cdot_i = \hat{\Xi}^I_{i,J}$ for the prospective view (3.15).

Theorem 12.11 (Display (3.16)). *The aggregated prospective estimator exceeds the aggregated retrospective one by $\sum_i \widehat{\mathrm{Var}}(\mathrm{CDR}_i(I+1) \mid \mathcal{D}_I) + 2\sum_{k>i>0} \hat{C}^I_{i,J}\hat{C}^I_{k,J}(\hat{\Xi}^I_{i,J} - \hat{\Phi}^I_{i,J})$: the same decoupling for aggregated accident years as for a single one.*

Proof. Termwise, Lemma 12.6 and ring algebra. □

Chapter 13

The observed data: Mack's hypotheses discharged

13.1 From independence across accident years to the hypotheses

Theorem 5.23 conditions on a σ-algebra $\mathcal{D}$ of observed data and carries two named hypotheses: that $\mathcal{D}$ says nothing about accident year i beyond that year's own history up to its latest observed development year. Theorem 8.17 carries a third, Definition 8.12. Mack obtains all three from assumption (M2) in the sentence "because of the independence of the accident years" in the proof of Theorem 3 [1, pp. 217–219]. This section names the σ-algebra and proves that sentence.

Definition 13.1 (The observed data)**.** After n calendar years the triangle holds one diagonal per accident year: row i is observed up to development year $n-1-i$. So

$$\mathrm{obsSigma} = \bigvee_{i<n} \sigma(C_{i,0}, \ldots, C_{i,n-1-i}),$$

and $\mathrm{otherObs}_i$ is the same join over the accident years $i' \neq i$.

Lemma 13.2 (The observed data splits at one accident year)**.** *For $i < n$, $\mathrm{obsSigma} = \sigma(C_{i,0}, \ldots, C_{i,n-1-i}) \vee \mathrm{otherObs}_i$, and $\mathrm{otherObs}_i$ is contained in the σ-algebra of all the other accident years.*

Proof. Both inclusions termwise: a term of the join either is row i's or is one of the others'. This is Lemma 5.29 for the latest diagonal instead of a development year. □

Lemma 13.3 (The chain-ladder estimators are functions of the observed data)**.** *S_k, $\hat{f}_k$ and $\hat{C}_{i,d+m}$ are obsSigma-measurable, with no hypothesis: an accident year contributes to $\hat{f}_k$ only when $C_{i,k+1}$ lies on or above the latest diagonal. This is the hypothesis $\hat{C}_{i,d+m}$ is $\mathcal{D}$-measurable of Theorem 5.23.*

Proof. $i \in \mathrm{contributors}(n,k)$ gives $i+k+2 \leq n$, hence $k+1 \leq n-1-i$; then induction on m for the product. □

Theorem 13.4 (No further information about one accident year)**.** *Assume (M2). For g integrable and measurable for accident year i, and $i < n$,*

$$E[g \mid \mathrm{obsSigma}] = E[g \mid \sigma(C_{i,0}, \ldots, C_{i,n-1-i})] = E[g \mid \mathcal{D}_{n-1-i}].$$

Taking $g = C_{i,k}$ and $g = C_{i,k}^2$ gives exactly the two "no extra information" hypotheses of Theorem 5.23.

Proof. Split the conditioning σ-algebra by Lemma 13.2 (for the first equality) and by Lemma 5.29 (for the second), and apply Theorem 5.28 with m_1 row i's observed history, m_1' the whole of row i and m_2 the other rows, which (M2) makes independent of m_1'. Both sides reduce to the same conditional expectation. □

Theorem 13.5 (Mack's Theorem 3 from Mack's own assumptions)**.** *Assume (M1row), (M3row), (M2), that $\mathcal{D}_k$ is generated by the rows' histories, and square integrability of the claims and of the chain-ladder prediction. Then, conditionally on the observed data,*

$$E\big[(\hat{C}_{i,d+m} - C_{i,d+m})^2 \mid \text{obsSigma}\big] = V_m + \big(\hat{C}_{i,d+m} - C_{i,d} \textstyle\prod_{l<m} f_{d+l}\big)^2, \qquad d = n-1-i.$$

Nothing is left over: the two hypotheses of Theorem 5.23 are Theorem 13.4, the measurability of the predictor is Lemma 13.3, and (M1), (M3) in the $\mathcal{D}_k$ form are Theorem 5.30.

Proof. Substitute those four into Theorem 5.23. □

Theorem 13.6 (The cross-term condition from independence)**.** *Assume (M2) and square integrability of the claims. For $i \neq j$ with $i < n$,*

$$E\big[(C_i - E[C_i \mid \text{obsSigma}])(C_j - E[C_j \mid \text{obsSigma}]) \bigm| \text{obsSigma}\big] = 0,$$

so Definition 8.12 holds for the true ultimates over all n accident years.

Proof. The argument of Theorem 5.31. Condition on the observed data together with the whole of accident year j: row j's centred ultimate is measurable there and comes out of the conditional expectation. What remains is row i's centred ultimate conditioned on a σ-algebra that adds nothing about row i (Theorem 5.28), and the centring was by exactly that conditional expectation (Theorem 13.4), so it vanishes. The tower property brings the identity back to the observed data. □

Theorem 13.7 (Mack's Corollary from Mack's own assumptions)**.** *Under the same hypotheses as Theorem 13.5, taken over all n accident years,*

$$E\Big[\Big(\sum_i \hat{C}_i - \sum_i C_i\Big)^2 \Bigm| \text{obsSigma}\Big] = \sum_i \text{procVar}_i + \Big(\sum_i (\hat{C}_i - M_i)\Big)^2 \ \geq\ \sum_i \text{procVar}_i.$$

The chain from Mack's row-conditioned assumptions plus independence across accident years to the exact Theorem 3 and its total-reserve form is closed.

Proof. Theorem 8.17 with $\mathcal{D} = \text{obsSigma}$, its cross-term hypothesis supplied by Theorem 13.6 and its remaining hypotheses by Theorem 13.4 and Lemma 13.3. □

Chapter 14

Mack's published numerical example

Definition 14.1 (Taylor-Ashe cumulative triangle)**.** The 55 observed cumulative claims in Mack's Table 1 are represented exactly as integers. Unobserved cells are zero and do not enter any estimator.

Theorem 14.2 (Published development factors)**.** *The nine development factors computed from Table 1 equal exact rational numbers whose decimal roundings are the values printed by [1].*

Proof. Each equality follows by normalization of the finite sums and division. □

Theorem 14.3 (Published estimable variances)**.** *The eight variance values supplied by Mack's variance estimator equal exact rationals with the printed decimal roundings. The ninth value is specified by an extrapolation convention and is not asserted here as an estimator value.*

Proof. Each equality follows by normalization of the weighted squared deviations. □

Definition 14.4 (Total reserve for Table 1)**.** The total reserve is the sum of the ten chain-ladder reserves computed from the published triangle.

Theorem 14.5 (Published total reserve)**.** *The exact total reserve is strictly between 18,680.5 and 18,681.5 thousand, so it rounds to the 18,681 thousand printed in Mack's Table 2.*

Proof. Normalization of the ten reserve terms proves both strict inequalities. □

Chapter 15

Non-vacuity: a nondegenerate model

Theorem 15.1 (A nondegenerate Mack model exists). *There is a probability space, a random triangle with three accident years, and constants f_k, σ_k^2 with $\sigma_0^2 > 0$, satisfying (M1), (M3) and (M2'), in which the first development step is genuinely random. The model: eight equally likely outcomes, an independent ± 1 shock per accident year, $C_{i,0} = c$ and $C_{i,k} = cf^k(1 + s\,\xi_i)$ for $k \geq 1$, with $\mathcal{D}_0$ trivial and $\mathcal{D}_k$ everything for $k \geq 1$.*

Proof. Conditional expectations given the trivial σ-algebra are eight-term averages; the shocks sum to zero, have unit square, and are orthogonal. □

Theorem 15.2 (Headline theorems instantiated). *On that model, $E[\hat{f}_0 \mid \mathcal{D}_0] = 2$ although $\hat{f}_0$ is random; $E[\hat{C}_{2,2} \mid \mathcal{D}_0] = E[C_{2,2} \mid \mathcal{D}_0]$; $E[(\hat{f}_0 - 2)^2 \mid \mathcal{D}_0] = \sigma_0^2/S_0$; and $E[\hat{\sigma}_0^2 \mid \mathcal{D}_0] = \sigma_0^2 = 4$.*

Proof. Direct instantiation of the general theorems; every hypothesis is discharged. □

Chapter 16

Reference computation

Mack's 1993 example (the Taylor and Ashe 1983 triangle) is reproduced to every printed digit (nine reserves, the total 18,681 thousand, all standard-error percentages) by a dependency-free script implementing the definitions above in floating point. The Lean definitions are over $\mathbb{R}$ and noncomputable; the script is the reference computation, not the Lean code.

Bibliography

[1] T. Mack, Distribution-free calculation of the standard error of chain ladder reserve estimates, ASTIN Bulletin 23 (1993) 213-225.

[2] T. Mack, The standard error of chain ladder reserve estimates: recursive calculation and inclusion of a tail factor, ASTIN Bulletin 29 (1999) 361-366.